



Stagnone di Marsala digital twin: project resumption and status

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1. Summary

Work on the coupled three-dimensional model resumed on 5 August after a pause since mid-June, due to the rewriting of the manuscripts related to GEE. As we already had several runs done and the skeleton of the paper in place, the main outcome of these days was to compile everything and finalise the first draft, in LaTeX and formatted for *Estuarine, Coastal and Shelf Science*.

The experimental design is a complete factorial in three factors, which needed eight simulations. Two of them, those combining a mobile bed with no wave coupling, required a change to the sediment configuration before they would run to completion, and that change had to be applied to the other mobile-bed members as well so that all four remain comparable. Those re-runs finished today, and the results below are from the complete set.

The manuscript is provisionally titled “**Interacting effects of wave coupling and bed mobility on Lagrangian transport in a shallow vegetated lagoon**”, with the three of us as authors. I would be grateful for your comments on both the framing and the content.

2. What the study asks and what it found

Coastal lagoon models are routinely validated against water level and then used to describe transport, so I set the paper to ask whether the first licenses the second. Eight configurations were run over the same nine days of July 2025, covering the drifter deployments, from the same initial condition and forcing, differing only in wave coupling, bed mobility, and whether seagrass resistance is uniform or distributed from a satellite classification. Each was scored against three tide gauges and against 35 GPS drifters over twelve deployments.

Water level does not separate the configurations. Anomaly RMSE differs between them by at most 4.2 mm at any station, within the measurement uncertainty, and no configuration is consistently best. Over the same simulations Liu–Weisberg skill spans 0.469 to 0.705 and endpoint separation 336 to 625 m. A calibration on water level alone would have had no basis on which to prefer any of them.

Neither wave coupling nor bed mobility improves Lagrangian skill on its own, and each is what makes the other work. Applied singly they are the two treatments that make the model worse. Applied together they give the ensemble its two best members. The two worst and the two best configurations are built from the same pair of treatments and differ only in whether both are switched on.

Change in Liu–Weisberg skill	On a fixed bed	On a mobile bed
From adding wave coupling	−0.035 to −0.042	+0.155 to +0.232
	Without waves	With waves
From adding bed mobility	−0.071 to −0.075	+0.119 to +0.199

Bed mobility without waves makes every one of the 35 drifters worse, in both roughness treatments, which is as consistent as a paired comparison can be. All eight bootstrap intervals behind the table are clear of zero.

Distributed seagrass roughness is a third-order refinement on this evidence. It changes skill by 0.009 or less in three of its four conditions, and by 0.081 in the fourth, which is on a mobile bed



with wave coupling. Its effect on the wave field is nonetheless real and correctly ordered by canopy density, reducing near-bed orbital velocity by 14.3% over *Posidonia* against 3.3% over exposed rock, so the parametrisation works. It simply does not move surface transport unless the other two are present.

Splitting the trajectory error into a magnitude part and a direction part shows the two moving together rather than independently, so the treatments appear to act on the velocity field as a whole rather than on separable parts of the error. And surface current speed exceeds bed speed by a factor of 1.85 to 2.20 at sub-metre mean depth, which is the quantitative case for solving in three dimensions, consistent with De Marchis et al. (2012) here. That ratio is set by wave coupling rather than by bed mobility.

3. Two questions for you

First, the title. The one in the draft is my current preference, but feel free to suggest changes. Here are other two alternatives that I think are defensible:

1. *Interacting effects of wave coupling and bed mobility on Lagrangian transport in a shallow vegetated lagoon.* In the draft. Conservative and conventional in form.
2. *Wave coupling helps or hurts lagoon transport prediction depending on whether the bed can move.* A bit more bold, try to state the finding rather than the topic, and the conditional form claims what the eight configurations show.
3. *Transport validation of a shallow lagoon model: interacting effects of wave coupling and bed mobility.* States the method first, then the mechanism.

Second, and more consequential, whether to submit before obtaining a new bathymetric survey. The configurations that best reproduce observed transport are the ones with a mobile bed, and its critical shear stress for erosion has not been constrained against measured bathymetric change. We are using the old bathymetry, so it is possible that a mobile bed helps because it moves the old morphology towards the present one, in which case the improvement is a compensation rather than a physical result. The manuscript states this openly, and notes that the lagoon bed changes by up to 1.4 m over nine days in isolated cells, which is not a plausible rate. If there is a realistic chance of obtaining new bathymetry soon, maybe we should wait for it, and rerun. If not, we can submit valuing to declare a caveat that the improvement may be a compensation for an outdated bathymetry.

4. Status and plan

The draft is complete across all six sections, with eight figures and three tables, 5650 words against a limit of 8000, and 25 references. Highlights and a cover letter are drafted. Outstanding is a graphical abstract, which *ECSS* requires.

After the holidays, and before leaving for Malta, my intention is to close this paper and begin drafting the remaining themes that can be made from this model, like the Isola Grande channel reopening scenario; wave-driven sediment mobilisation with the Trapani canal contribution to lagoon siltation, etc.

The full draft, figures and analysis code are in the project repository, and I am sending together here the compiled PDF.

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